The Connes–Størmer entropy of two maximal abelian subalgebras of M_2: an explicit variational formula, and partitions that beat bases in M_n

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Abstract

The Connes–Størmer relative entropy $H(P|Q)$ measures the relative position of two subalgebras P, Q of a finite von Neumann algebra. For the diagonal masa Δ_n of $M_n(\mathbb{C})$ and a rotated copy $U\Delta_n U^*$, Choda's value $h(U)$, the averaged row entropy of the unistochastic matrix $(|u_{ij}|^2)$, is a lower bound; Petz, Szántó and Weiner showed that it is not the entropy already for $n = 2$, and a formula has remained open even in that case. We prove that for $U \in U(2)$ with $|u_{11}|^2 = \cos^2\theta$ the entropy is the largest integral of $\psi(\beta) = \sin 2\beta \log\tan^2\beta$ over a window of length θ, and that it is attained by an orthonormal basis from the one-parameter family used by Petz, Szántó and Weiner, whose angle we locate. The mechanism is the antipodal symmetry of the Bloch ball, which forces the entropy to equal the maximum of an entropy difference over single states. Consequently $H = \kappa\theta + O(\theta^3)$ with $\kappa = \max\psi \approx 1.3255$, whereas $h(U) \sim 2\theta^2 \log\frac{1}{\theta}$. For $n \geq 3$ the symmetry is lost. We give a first-order criterion under which Choda's partition is not optimal, valid on an open set of unitaries in every dimension, and a numerical study for $n = 3, 4, 5$: for a small fraction of Haar-random U the best partitions found have $n+1$ rank-one elements and beat every orthonormal basis found.

Keywords: Connes–Størmer entropy; relative position of subalgebras; maximal abelian subalgebras; unistochastic matrices; mutually unbiased bases; entropic uncertainty; partitions of unity.
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1 Introduction

1.1 The relative position of two subalgebras

A pair of subspaces of a finite-dimensional Hilbert space is determined up to unitary equivalence by their dimensions and the principal angles between them. For two subalgebras of a von Neumann algebra no such complete invariant is available, and a number of numerical invariants have been proposed to measure how far the two algebras are from each other: the Jones index and the Pimsner–Popa probabilistic index for inclusions [10, 19], Popa's notion of orthogonal pairs [20], angles and distances between subalgebras (see the recent work of Bakshi and Guin [1] and of Ghosh and Kumar [7]), and the relative entropy of Connes and Størmer [5], the subject of this paper.

Let $\mathcal{M}$ be a finite von Neumann algebra with a faithful normal tracial state τ, let $\eta(t) = -t\log t$, and for a von Neumann subalgebra $P \subset \mathcal{M}$ let E_P denote the τ-preserving conditional expectation onto P. The Connes–Størmer relative entropy of two subalgebras P, Q is

$$H(P|Q) = \sup \sum_j \big[\tau\eta(E_Q x_j) - \tau\eta(E_P x_j)\big], \tag{1}$$

the supremum being taken over all finite partitions of unity $x_j \geq 0$, $\sum_j x_j = 1$, in $\mathcal{M}$. Each term compares how much a positive element x_j is "smeared out" by the two conditional expectations: if

x_j lies in P, then $E_P x_j = x_j$ while $E_Q x_j$ may be much more mixed. The trivial partition $x_1 = 1$ shows $H(P|Q) \geq 0$, and $H(P|P) = 0$. If $P \subset Q$, every term is non-positive by the concavity of η, so $H(P|Q) = 0$; $H(P|Q)$ is therefore a measure of the part of P that is not contained in Q. Connes and Størmer introduced (1) for finite-dimensional subalgebras as the basic ingredient of their entropy of automorphisms [5]; Pimsner and Popa extended it to arbitrary subalgebras and related the entropy $H(\mathcal{M}|N)$ of an inclusion of type II_1 factors to the Jones index and to their probabilistic index [19, 10].

1.2 Two maximal abelian subalgebras

The simplest pair of subalgebras for which (1) is not trivial is a pair of maximal abelian subalgebras (masas) of $M_n = M_n(\mathbb{C})$ with $\tau = \mathrm{Tr}\,/n$: the diagonal algebra Δ_n and a rotated copy $U\Delta_n U^*$, U unitary. In the language of quantum mechanics these are two non-degenerate observables, or two orthonormal bases of $\mathbb{C}^n$: the standard basis and the columns of U. Choosing the partition into the minimal projections e_{jj} of Δ_n gives the lower bound

$$H(\Delta_n|U\Delta_n U^*) \;\geq\; h(U) = \frac{1}{n}\sum_{i,j} \eta\big(|u_{ij}|^2\big), \tag{2}$$

the Shannon entropy of the rows of the unistochastic matrix $(|u_{ij}|^2)$, averaged over the rows. Choda [3] showed that $h(U)$ is the value of a modified relative entropy in which the partitions are restricted to Δ_n, and it was expected that $h(U)$ is also the value of H. At one extreme this is correct: if the two masas are complementary (equivalently, $|u_{ij}|^2 = \frac{1}{n}$ for all i, j, i.e. $\sqrt{n}\,U$ is a complex Hadamard matrix and the two bases are mutually unbiased), then $h(U) = \log n$, which is the largest possible value of H, and Petz, Szántó and Weiner proved that $H = \log n$ holds only for complementary masas [18, Theorem 3]. Away from this extreme, however, they showed that already for $n = 2$ the inequality (2) is strict in general, and remarked that the correction of the formula "seems to be a hard problem" [18, Appendix]. Bakshi and Guin, who need lower bounds for the entropy of pairs of spin model subfactors built from 2×2 complex Hadamard matrices, use $h(U)$ and note that a formula for $H(\Delta_n|U\Delta_n U^*)$ is open even for $n = 2$ [1, Section 2.3 and Theorem 6.22].

What makes the problem hard is the nature of the supremum in (1). It is taken over partitions of arbitrary length whose elements need not commute with each other or with either algebra, and the functional is neither convex nor concave in the partition. Petz, Szántó and Weiner rewrote it as a maximisation over ensembles of states with barycentre $\frac{1}{n}$ (Proposition 2.3): with

$$G(\rho) = S(E_Q\rho) - S(E_P\rho)$$

for a density matrix ρ and the von Neumann entropy S, $H(P|Q)$ is the largest average of G over ensembles $\{t_j, \rho_j\}$ with $\sum_j t_j\rho_j = \frac{1}{n}$. Geometrically, $H(P|Q)$ is the value at the centre $\frac{1}{n}$ of the concave envelope of G on the state space, and in particular $H(P|Q) \leq G^* := \max G$ (Proposition 3.1).

1.3 Results for $n = 2$: why the qubit case is solvable

For $n = 2$ the state space is the Bloch ball, and the observation that unlocks the problem is that the antipodal map $r \mapsto -r$, i.e. $\rho \mapsto 1 - \rho$, preserves G for every pair of subalgebras of M_2: a 2×2 density matrix and its reflection have the same spectrum up to order, and conditional expectations are unital. A maximiser ρ^* of G and its antipode $1 - \rho^*$ average to $\frac{1}{2}$, so the concave envelope at the centre is just the maximum, $H = G^*$, and an optimal partition is $\{\rho^*, 1 - \rho^*\}$ (Lemma 3.2). For two masas we show further that G is maximised at a pure state in the great circle spanned by the Bloch vectors of the two bases, so the optimum is an orthonormal basis, and the maximisation over that

circle reduces to a one-dimensional problem. To state the result, write $h_2(p) = \eta(p) + \eta(1-p)$ for the binary entropy and

$$\psi(\beta) = \sin(2\beta)\log\tan^2\beta \qquad (\beta \in \mathbb{R}), \tag{3}$$

extended by continuity with $\psi = 0$ on $\frac{\pi}{2}\mathbb{Z}$. Every unitary $U \in M_2$ satisfies $|u_{11}| = |u_{22}| = \cos\theta$ for a unique $\theta \in [0, \frac{\pi}{2}]$, the angle between the lines $\mathbb{C}e_1$ and $\mathbb{C}Ue_1$; Choda's value is $h(U) = h_2(\cos^2\theta)$.

Theorem A. *Let $U \in M_2$ be unitary, $c = |u_{11}|^2 = \cos^2\theta$ with $\theta \in [0, \frac{\pi}{2}]$, and*

$$\varphi(\alpha) = h_2\big(\cos^2(\alpha-\theta)\big) - h_2(\cos^2\alpha) = \int_{\alpha-\theta}^{\alpha} \psi(\beta)\,d\beta.$$

Then $H(\Delta_2|U\Delta_2U^*) = \max_{\alpha\in\mathbb{R}}\varphi(\alpha) = G^*$. The maximum is attained by the partition into the two projections onto an explicit orthonormal basis of $\mathbb{C}^2$, determined by a maximiser α^* of φ and the phases of U. For $\frac{1}{2} < c < 1$ the maximiser is unique modulo $\frac{\pi}{2}$, lies in $(\theta - \frac{\pi}{4}, 0)$ and is the unique solution there of $\psi(\alpha) = \psi(\alpha-\theta)$; H is invariant under $c \mapsto 1-c$ (Theorem 4.4 and Proposition 4.6).

Thus $\varphi(\alpha)$ is the value of the basis at angle α, and it is the area under ψ over the window $[\alpha-\theta, \alpha]$. Choda's partition is the window $[-\theta, 0]$; the optimal basis slides the window so that it captures the hump of ψ on $(-\frac{\pi}{4}, 0)$ (Figure 2). The optimal basis lies on the other side of the standard basis than the basis of $U\Delta_2U^*$ (Figure 1): it gives up some entropy of $E_{\Delta_2}x_j$ in exchange for more entropy of $E_{U\Delta_2U^*}x_j$. Integrating $\psi \le \kappa := \max\psi$ over the window gives a bound linear in the angle, and a Taylor expansion at the maximum of ψ shows that it is sharp to first order.

Theorem B. *With the notation of Theorem A, $h(U) \le H(\Delta_2|U\Delta_2U^*) \le \kappa\min(\theta, \frac{\pi}{2} - \theta)$, where $\kappa = 2\sqrt{1-y^2}/y \approx 1.3255$ and $y \in (0,1)$ is the unique solution of $y\operatorname{artanh} y = 1$. As $\theta \to 0$,*

$$H(\Delta_2|U\Delta_2U^*) = \kappa\theta - \frac{\kappa + 4/\kappa}{6}\,\theta^3 + O(\theta^4), \qquad h(U) = 2\theta^2\log\frac{1}{\theta} + \theta^2 + O\Big(\theta^4\log\frac{1}{\theta}\Big)$$

(Theorem 5.1).

The qualitative message is that for two nearby masas the Connes–Størmer entropy is of *first* order in the angle between them, while Choda's value is of second order up to a logarithm; in particular $H/h(U) \to \infty$ as $\theta \to 0$. Restricting the partitions to one of the two algebras, as in Choda's modified entropy, loses the leading-order term entirely (Figure 3).

Relation to the work of Petz, Szántó and Weiner. In the appendix of [18] the authors consider, for Δ_2 and the masa generated by $\sin\beta\,\sigma_1 + \cos\beta\,\sigma_3$, the one-parameter family of partitions $\{r(t), 1-r(t)\}$ with $r(t) = \frac{1}{2}(1 + \sin t\,\sigma_1 + \cos t\,\sigma_3)$, and the resulting lower bound $f(t) \le H$; they show that $f'(0) \neq 0$ in general, so that Choda's partition $t = 0$ is not optimal (Corollary 5.2). In our normal form $\beta = 2\theta$ and $f(t) = \varphi(t/2)$. Theorem A therefore says precisely that the supremum over *all* partitions is attained inside their family, $H = \max_t f(t)$. What is new is the proof that no other partition does better, whatever the number of its elements, their ranks and the position of their Bloch vectors, and the location of the optimal parameter; the family itself, the function f and the non-optimality of $t = 0$ are theirs.

1.4 Results for $n \ge 3$

For $n \ge 3$ the mechanism breaks down. The natural analogue of the antipodal map is the point reflection $\rho \mapsto \frac{2}{n} - \rho$ through the centre of the state space, but it maps a pure state to an operator with the negative eigenvalue $\frac{2}{n} - 1$; there is no symmetry pairing maximisers of G around $\frac{1}{n}$. Two things

survive. First, a first-order perturbation of Choda's partition shows $H(\Delta_n|U\Delta_nU^*) > h(U)$ whenever a certain matrix built from U, $M_U = (u_{jk}\log|u_{jk}|^2)_{jk}\,U^*$, is not Hermitian (Proposition 6.1); this holds on an open set of unitaries in every dimension and recovers the result of Petz, Szántó and Weiner for $n = 2$. Second, we computed $H(\Delta_n|U\Delta_nU^*)$ numerically for Haar-random U with $n = 3, 4, 5$, maximising separately over general partitions and over orthonormal bases (Section 6). The picture that emerges differs from $n = 2$ in three respects. For most unitaries the best partition found is an orthonormal basis, but for a small fraction of them (about 7% for $n = 3$) the best partitions found consist of $n + 1$ rank-one elements and beat every basis found, by up to 6.9×10^{-3}. In every sample the best value found lies strictly between $h(U)$ and the computed G^*, so that neither Choda's lower bound nor the envelope bound appears to be attained. These are numerical observations: every value we report is attained at an explicit partition and recomputed with 50 digits, hence it is a rigorous lower bound for H up to rounding, but the maxima over all bases are not certified. To turn the first observation into a theorem it would suffice to certify a global upper bound for the best basis for a single unitary, a global optimisation problem over a six-dimensional flag manifold for $n = 3$; by continuity the strict inequality would then hold on a set of positive Haar measure (Section 6.3).

1.5 Related work

Entropy of subalgebras. The relative entropy (1) was introduced by Connes and Størmer [5] and extended to arbitrary subalgebras, with the connection to index theory, by Pimsner and Popa [19]; the monographs of Ohya and Petz [15] and of Neshveyev and Størmer [14] treat it as part of the theory of quantum and dynamical entropy. For finite-dimensional algebras the reformulation in terms of ensembles of states, which is our starting point, is due to Petz, Szántó and Weiner [18].

Relative position and complementarity. Popa [20] introduced orthogonal pairs of subalgebras of a finite von Neumann algebra, $\tau(xy) = \tau(x)\tau(y)$ for $x \in P$, $y \in Q$; for masas of M_n this is the complementary case $|u_{ij}|^2 = \frac{1}{n}$, and Choda showed that $h(U) = \log n$ exactly for orthogonal pairs [3]. Complementary subalgebras are the algebraic form of complementary observables in quantum mechanics [17]; for masas they are pairs of mutually unbiased bases [25, 6], and the corresponding unitaries are rescaled complex Hadamard matrices [22]. The matrices $(|u_{ij}|^2)$ that appear in (2) are the unistochastic matrices, whose geometry inside the Birkhoff polytope is intricate already for $n = 3$ [2]. Bakshi and Guin [1] compute a list of invariants of relative position (the Pimsner–Popa constant, angles, and bounds for the Connes–Størmer entropy) for pairs of spin model subfactors, and Ghosh and Kumar [7] relate the Kadison–Kastler distance and the interior angle of two masas to Choda's entropy and to Popa's probabilistic index.

The quantum-information reading. Let $P = \Delta_n$ and $Q = U\Delta_nU^*$. For an ensemble $\{t_j, \rho_j\}$ with barycentre $\frac{1}{n}$, let J be the label (drawn with probabilities t_j) and X_P, X_Q the outcomes of measuring the state ρ_J in the standard basis and in the basis of columns of U. Then

$$\sum_j t_j G(\rho_j) = I(J : X_P) - I(J : X_Q)$$

with the classical mutual information I (Proposition 2.4). Hence $H(\Delta_n|U\Delta_nU^*)$ is the largest amount by which the first measurement can be more informative than the second about the preparation of an ensemble whose average is maximally mixed. Choda's partition, the ensemble of standard basis vectors, has $I(J : X_P) = \log n$ and $I(J : X_Q) = \log n - h(U)$. Entropic uncertainty relations, beginning with Maassen and Uffink [12], bound the sum $S(E_P\rho) + S(E_Q\rho)$ from below, and averaged over an ensemble they yield Hall's information exclusion principle, an upper bound for $I(J : X_P) +$

$I(J : X_Q)$ [8]; see the review of Coles, Berta, Tomamichel and Wehner [4]. The Connes–Størmer entropy asks the complementary question about the *difference* of the two informations, under the constraint that the ensemble average is $\frac{1}{n}$.

What is new. The ensemble form of H (Proposition 2.3), the bound $H \le \log n$, the strict inequality $H(\Delta_2|U\Delta_2U^*) > h(U)$ for $c \notin \{0, \frac{1}{2}, 1\}$, the one-parameter family of bases in which our optimum lies, and the characterisation of $H = \log n$ are due to Petz, Szántó and Weiner [18]; the lower bound (2) and its interpretation are Choda's [3]. Theorem A, Theorem B, the antipodal argument of Lemma 3.2, Proposition 6.1 and the numerical findings of Section 6 are new to the best of our knowledge. The envelope bound of Proposition 3.1 is an elementary consequence of the ensemble form, and Proposition 2.4 is a direct rewriting of it; we include both because they organise the argument and separate $n = 2$ from $n \ge 3$. A short account of part of these results appears as a use case in the description of the software used for the computations [11].

Organisation. Section 2 collects definitions, the ensemble form and its information-theoretic reading. Section 3 proves the envelope bound and the antipodal lemma. Section 4 proves Theorem A and locates the optimal angle, and Section 5 proves Theorem B. Section 6 treats $n \ge 3$: the first-order criterion, the numerical findings, and what is needed to make them rigorous. Section 7 lists open problems, and Appendix A describes the numerical method.

2 Preliminaries

2.1 Conditional expectations and the entropy

Throughout, $\mathcal{M} = M_n$, $\tau = \operatorname{Tr}/n$, and $\langle x, y\rangle_\tau = \tau(x^*y)$. Subalgebras are unital $*$-subalgebras. For such a subalgebra P the τ-preserving conditional expectation E_P is the orthogonal projection of $(\mathcal{M}, \langle\cdot,\cdot\rangle_\tau)$ onto P; it is unital, positive, P-bimodular, preserves τ and Tr, and is the unique conditional expectation onto P with these properties (see e.g. [23]). The function $\eta(t) = -t\log t$, $\eta(0) = 0$, is continuous and concave on $[0,\infty)$; for $x \ge 0$, $\eta(x)$ is defined by functional calculus, and $S(\sigma) = \operatorname{Tr}\eta(\sigma)$ is the von Neumann entropy of a density matrix σ. The state space $\mathcal{D} = \{\rho \in M_n : \rho \ge 0,\ \operatorname{Tr}\rho = 1\}$ is a compact convex set of real dimension $n^2 - 1$. We write $h_2(p) = \eta(p) + \eta(1-p)$ for the binary entropy, $p \in [0,1]$, and reserve the letter h for Choda's value (2), which we regard as a function on all unitaries.

Definition 2.1. For subalgebras $P, Q \subset \mathcal{M}$ and $x \ge 0$ put $F(x) = F_{P,Q}(x) = \tau\eta(E_Qx) - \tau\eta(E_Px)$. The Connes–Størmer relative entropy is $H(P|Q) = \sup\sum_j F(x_j)$ over all finite families $x_1, \dots, x_m \ge 0$ in $\mathcal{M}$ with $\sum_j x_j = 1$ (partitions of unity).

Lemma 2.2. *Let $P, Q \subset \mathcal{M}$ be subalgebras.*

(a) $F(\lambda x) = \lambda F(x)$ for $x \ge 0$ and $\lambda \ge 0$. In particular, merging two elements of a partition that are multiples of the same positive operator does not change $\sum_j F(x_j)$.

(b) Let $v \in \mathcal{M}$ be unitary and $\gamma = \operatorname{Ad} v$. Then $E_{\gamma(B)} = \gamma \circ E_B \circ \gamma^{-1}$ for every subalgebra B, $F_{\gamma(P),\gamma(Q)}(\gamma(x)) = F_{P,Q}(x)$, and $H(\gamma(P)|\gamma(Q)) = H(P|Q)$; γ maps optimal partitions for (P, Q) onto optimal partitions for $(\gamma(P), \gamma(Q))$.

Proof. (a) For $s, \lambda \ge 0$, $\eta(\lambda s) = \lambda\eta(s) + s\,\eta(\lambda)$, hence $\tau\eta(\lambda y) = \lambda\tau\eta(y) + \eta(\lambda)\tau(y)$ for $y \ge 0$. Since $\tau(E_Qx) = \tau(E_Px) = \tau(x)$, the terms with $\eta(\lambda)$ cancel in $F(\lambda x)$. If $x_1 = a\,y$ and $x_2 = b\,y$ with $a, b \ge 0$, then $F(x_1) + F(x_2) = (a+b)F(y) = F(x_1 + x_2)$.

(b) γ is unitary for $\langle\cdot,\cdot\rangle_\tau$, since $\tau(\gamma(x)^*\gamma(y)) = \tau(vx^*yv^*) = \tau(x^*y)$, and $\gamma(B)$ is the image of B; so $\gamma E_B \gamma^{-1}$ is the orthogonal projection onto $\gamma(B)$. Moreover $\eta(\gamma(y)) = \gamma(\eta(y))$ and $\tau \circ \gamma = \tau$, so $\tau\eta(E_{\gamma(B)}\gamma(x)) = \tau\gamma(\eta(E_B x)) = \tau\eta(E_B x)$. Since γ maps partitions of unity bijectively onto partitions of unity, the remaining claims follow. □

2.2 Two masas

Let Δ_n be the diagonal masa and $Q_U = U\Delta_n U^*$ for a unitary U with columns $u_1, \dots, u_n$. The minimal projections of Q_U are $p_k = u_k u_k^*$, and

$$E_{\Delta_n}(x) = \sum_k e_{kk} x e_{kk}, \qquad E_{Q_U}(x) = \sum_k p_k x p_k = \sum_k \langle u_k, x u_k\rangle\, p_k, \tag{4}$$

since $\sum_k p_k x p_k \in Q_U$ and $\tau\big(p_l(x - \sum_k p_k x p_k)\big) = \tau(p_l x) - \tau(p_l x p_l) = 0$ for every l. Thus the spectrum of $E_{Q_U} x$ is the diagonal of $U^* x U$, and

$$F(x) = \frac{1}{n}\Big[\sum_k \eta\big((U^*xU)_{kk}\big) - \sum_k \eta(x_{kk})\Big]. \tag{5}$$

The algebra Q_U does not change if U is multiplied from the right by a diagonal unitary or a permutation matrix, and by Lemma 2.2(b) with $\gamma = \operatorname{Ad} D$ for a diagonal unitary D (which fixes Δ_n), $H(\Delta_n|Q_{DU}) = H(\Delta_n|Q_U)$. For the partition $(e_{kk})_k$, $\eta(x_{kk}) \in \{\eta(0), \eta(1)\} = \{0\}$ and $(U^* e_{kk} U)_{jj} = |u_{kj}|^2$, which gives (2). For an orthonormal basis $w_1, \dots, w_n$, the columns of a unitary W, the partition $x_j = w_j w_j^*$ gives, by (5),

$$\sum_j F(w_j w_j^*) = h(U^*W) - h(W), \tag{6}$$

and Choda's partition is the case $W = 1$. We call (6) the value of the basis W.

2.3 The ensemble form

The following reformulation is due to Petz, Szántó and Weiner [18, Section 3, Lemma 1 and (7), (8)]; we include the short proof, and a Carathéodory argument for the attainment of the supremum.

Proposition 2.3 (Petz–Szántó–Weiner)**.** *Let $P, Q \subset M_n$ be subalgebras and $G(\rho) = S(E_Q\rho) - S(E_P\rho)$ for $\rho \in \mathcal{D}$.*

(a) A partition (x_j) with $x_j \neq 0$ corresponds to the ensemble $t_j = \tau(x_j)$, $\rho_j = x_j/\operatorname{Tr} x_j$, which satisfies $t_j > 0$, $\sum_j t_j = 1$ and $\sum_j t_j \rho_j = \frac{1}{n}$; conversely every such ensemble defines the partition $x_j = n t_j \rho_j$. In both directions $\sum_j F(x_j) = \sum_j t_j G(\rho_j)$.

(b) $H(P|Q) = \max\big\{\sum_j t_j G(\rho_j) :\ t_j \geq 0,\ \sum_j t_j = 1,\ \sum_j t_j \rho_j = \frac{1}{n}\big\}$; the maximum is attained by an ensemble of at most $n^2 + 1$ states. (Petz, Szántó and Weiner show that n^2 linearly independent states suffice.)

(c) $G \leq \log n$ on $\mathcal{D}$, hence $H(P|Q) \leq \log n$.

Proof. (a) Elements $x_j = 0$ contribute $F(0) = 0$ and can be dropped. $\operatorname{Tr} x_j = n\tau(x_j) = n t_j$, so $x_j = n t_j \rho_j$, and $\sum_j x_j = 1$ is equivalent to $\sum_j t_j \rho_j = \frac{1}{n}$, which implies $\sum_j t_j = 1$ by taking traces. Since $E_P\rho$ and $E_Q\rho$ are density matrices, $F(\rho) = \frac{1}{n}[S(E_Q\rho) - S(E_P\rho)] = \frac{1}{n}G(\rho)$, and Lemma 2.2(a) gives $F(x_j) = n t_j F(\rho_j) = t_j G(\rho_j)$.

(b) By (a), H is the supremum of $\sum_j t_j G(\rho_j)$ over finite ensembles with barycentre $\frac{1}{n}$. Identify the affine hull of $\mathcal{D}$ with $\mathbb{R}^{n^2-1}$. For such an ensemble the point $(\frac{1}{n}, \sum_j t_j G(\rho_j)) \in \mathbb{R}^{n^2}$ is a convex combination of points of the graph $\{(\rho, G(\rho)) : \rho \in \mathcal{D}\}$; by Carathéodory's theorem it is a convex combination of at most n^2+1 of them, i.e. there is an ensemble of at most n^2+1 states with the same barycentre and the same value. The ensembles of n^2+1 states (zero weights allowed) with barycentre $\frac{1}{n}$ form a compact set, on which $\sum_j t_j G(\rho_j)$ is continuous, because S is continuous on $\mathcal{D}$ and E_P, E_Q are linear. So the supremum is a maximum.

(c) $S(E_Q\rho) \le \log n$ for the density matrix $E_Q\rho$ of M_n, and $S(E_P\rho) \ge 0$. □

For two masas, $G(\rho) = H_{\mathrm{Sh}}\big(\mathrm{diag}(U^*\rho U)\big) - H_{\mathrm{Sh}}\big(\mathrm{diag}\,\rho\big)$ with the Shannon entropy H_{Sh} of a probability vector, by (4).

2.4 The information-theoretic reading

For an ensemble $\{t_j, \rho_j\}$ and a subalgebra $P \subset M_n$ let

$$\chi_P\{t_j,\rho_j\} = S\Big(\sum_j t_j E_P\rho_j\Big) - \sum_j t_j S(E_P\rho_j)$$

be the Holevo quantity of the ensemble $\{t_j, E_P\rho_j\}$, i.e. of the ensemble after the quantum channel E_P. If P is the masa of the minimal projections $p_1,\dots,p_n$, then $E_P\rho_j = \sum_k \mathrm{Tr}(p_k\rho_j)\,p_k$ by (4), and χ_P is the classical mutual information $I(J:X_P)$ between a label J with distribution (t_j) and the outcome X_P of the measurement (p_k) on ρ_J, whose conditional distribution is $\Pr(X_P = k \,|\, J = j) = \mathrm{Tr}(p_k\rho_j)$.

Proposition 2.4. *Let $P, Q \subset M_n$ be subalgebras and $\{t_j, \rho_j\}$ an ensemble with $\sum_j t_j\rho_j = \frac{1}{n}$. Then*

$$\sum_j t_j G(\rho_j) = \chi_P\{t_j,\rho_j\} - \chi_Q\{t_j,\rho_j\}, \qquad \textit{hence} \qquad H(P|Q) = \max\big(\chi_P - \chi_Q\big)$$

over these ensembles. For $P = \Delta_n$ and $Q = U\Delta_nU^*$ this reads $\sum_j t_j G(\rho_j) = I(J:X_P) - I(J:X_Q)$, where X_P and X_Q are the outcomes of measuring ρ_J in the standard basis and in the basis $u_1,\dots,u_n$.

Proof. E_P is linear and unital, so $\sum_j t_j E_P\rho_j = E_P(\frac{1}{n}) = \frac{1}{n}$ and $S(\sum_j t_j E_P\rho_j) = \log n$; the same holds for Q. Hence $\chi_P - \chi_Q = \sum_j t_j\big[S(E_Q\rho_j) - S(E_P\rho_j)\big] = \sum_j t_j G(\rho_j)$, and Proposition 2.3(b) gives the formula for H. The last statement is the identification $\chi_P = I(J:X_P)$ above. □

In this language Choda's partition is the ensemble of the standard basis vectors with equal weights, for which $X_P = J$, $I(J:X_P) = \log n$ and $I(J:X_Q) = \log n - h(U)$, and the complementary case is the one in which this ensemble is completely invisible to the second measurement. Since the Shannon entropy of a probability vector is at least the negative logarithm of its largest entry, $h(U) \ge -\log\max_{i,j}|u_{ij}|^2$; this is the bound of Maassen and Uffink [12] for $S(E_P\rho) + S(E_Q\rho)$, evaluated at the basis vectors $\rho = e_{kk}$, where $S(E_P\rho) = 0$.

3 The concave envelope and the antipodal symmetry

For a bounded function g on $\mathcal{D}$ let $\hat g(\rho) = \sup\{\sum_j t_j g(\rho_j) :$ finite ensembles with $\sum_j t_j\rho_j = \rho\}$; this is the smallest concave function on $\mathcal{D}$ that majorises g (its concave envelope).

Proposition 3.1. *Let $P, Q \subset M_n$ be subalgebras, G as in Proposition 2.3, $G^* = \max_{\mathcal{D}} G$ and $A = \{\rho \in \mathcal{D} : G(\rho) = G^*\}$. Then*

$$H(P|Q) = \hat G\big(\tfrac{1}{n}\big) \le G^*,$$

and $H(P|Q) = G^*$ if and only if $\frac{1}{n} \in \operatorname{conv} A$. In that case the optimal partitions are exactly the partitions $x_j = nt_j\rho_j$ with $\rho_j \in A$ whenever $t_j > 0$.

Proof. The identity $H = \hat{G}(\frac{1}{n})$ is Proposition 2.3(b) with the definition of $\hat{G}$. For every ensemble, $\sum_j t_j G(\rho_j) \le \sum_j t_j G^* = G^*$, with equality if and only if $G(\rho_j) = G^*$ whenever $t_j > 0$. If $\frac{1}{n} = \sum_j t_j\rho_j$ with $\rho_j \in A$, this ensemble has the value G^*, so $H = G^*$. Conversely, if $H = G^*$, the maximum in Proposition 2.3(b) is attained by an ensemble with value G^*, all of whose states with positive weight lie in A; hence $\frac{1}{n} \in \operatorname{conv} A$. The last statement is the equality case together with Proposition 2.3(a). □

The bound G^* is a maximisation over the (n^2-1)-dimensional state space, whereas H requires ensembles of up to n^2 states with a linear constraint; any state gives a lower bound for G^*, any partition a lower bound for H. Upper bounds for H come from the dual side: if $a(\rho) = a_0 + \operatorname{Tr}(A\rho)$ is an affine function with $a \ge G$ on $\mathcal{D}$, then $\sum_j t_j G(\rho_j) \le a(\sum_j t_j\rho_j)$ for every ensemble, and so

$$H(P|Q) \le a\big(\tfrac{1}{n}\big) = a_0 + \tfrac{1}{n}\operatorname{Tr} A. \tag{7}$$

Since $\hat{G}$ is concave and upper semicontinuous (the maximum in Proposition 2.3(b) is over a compact set), it is the infimum of its affine majorants [21, Section 12], so these bounds are sharp in the limit. The constant function $a = G^*$ gives Proposition 3.1.

The equality case of Proposition 3.1 is exactly what a symmetry of the state space can provide.

Lemma 3.2 (Antipodal symmetry)**.** *Let $n = 2$ and let $P, Q \subset M_2$ be subalgebras. The map $\rho \mapsto 1-\rho$ is an affine involution of $\mathcal{D}$ that fixes $\frac{1}{2}$ and satisfies $G(1-\rho) = G(\rho)$. Consequently $H(P|Q) = G^*$, and for every maximiser ρ^* of G the partition $\{\rho^*, 1-\rho^*\}$ is optimal. In the Bloch parametrisation $\rho = \frac{1}{2}(1 + r\cdot\sigma)$ the map is $r \mapsto -r$.*

Proof. If $\rho \ge 0$ and $\operatorname{Tr}\rho = 1$, the eigenvalues $\lambda, 1-\lambda \in [0,1]$ of ρ give $1-\rho \ge 0$, and $\operatorname{Tr}(1-\rho) = 1$. Since E_P is linear and unital, $E_P(1-\rho) = 1 - E_P\rho$, and for a 2×2 density matrix σ with eigenvalues $\mu, 1-\mu$ the matrix $1-\sigma$ has the eigenvalues $1-\mu, \mu$, so $S(1-\sigma) = S(\sigma)$. Hence $S(E_P(1-\rho)) = S(E_P\rho)$, likewise for Q, and $G(1-\rho) = G(\rho)$. If $\rho^* \in A$, then $1-\rho^* \in A$ and $\frac{1}{2} = \frac{1}{2}\rho^* + \frac{1}{2}(1-\rho^*) \in \operatorname{conv} A$, so $H = G^*$ by Proposition 3.1, with the optimal partition $x_\pm = 2\cdot\frac{1}{2}\rho_\pm$, $\rho_+ = \rho^*$, $\rho_- = 1-\rho^*$. □

Remark 3.3. (i) For $n \ge 3$ the point reflection $\rho \mapsto \frac{2}{n} - \rho$ through the centre of $\mathcal{D}$ is affine and fixes $\frac{1}{n}$, but it does not preserve $\mathcal{D}$: a pure state is mapped to an operator with the eigenvalue $\frac{2}{n} - 1 < 0$. Nothing forces $\frac{1}{n} \in \operatorname{conv} A$, and for $n = 3, 4, 5$ the computations of Section 6 find $G^* > H$ for every sampled unitary, i.e. the optimal ensembles do not consist of maximisers of G. (ii) If U is complementary, $G(e_{11}) = \log n = G^*$ by (5), and $\frac{1}{n} = \frac{1}{n}\sum_k e_{kk}$ is a convex combination of the maximisers e_{kk} (all of which satisfy $G(e_{kk}) = \log n$); so $H = G^* = \log n$, the partition (e_{kk}) is optimal, and $h(U) = \log n$.

4 Two masas of M_2

4.1 The function ψ

Let $\Psi(\beta) = h_2(\cos^2\beta)$ and let ψ be as in (3). Throughout, β denotes the variable of ψ and Ψ.

Lemma 4.1. *(a) ψ is continuous on $\mathbb{R}$, odd, of period $\frac{\pi}{2}$, vanishes exactly on $\frac{\pi}{4}\mathbb{Z}$, and $\psi > 0$ on $(-\frac{\pi}{4}, 0)$, $\psi < 0$ on $(0, \frac{\pi}{4})$.*

(b) Ψ is continuously differentiable on $\mathbb{R}$, $\Psi' = -\psi$, and Ψ is even and of period $\frac{\pi}{2}$.

(c) On $(-\frac{\pi}{4}, 0)$, ψ has a unique critical point β_κ; ψ is strictly increasing on $[-\frac{\pi}{4}, \beta_\kappa]$ and strictly decreasing on $[\beta_\kappa, 0]$. With $\beta_\kappa = -s^*/2$ and $y = \cos s^*$, $y \in (0,1)$ is the unique solution of $y \operatorname{artanh} y = 1$, and

$$\kappa := \max_{\mathbb{R}} \psi = \psi(\beta_\kappa) = 2 \tan s^* = \frac{2\sqrt{1-y^2}}{y}, \qquad \psi''(\beta_\kappa) = -4\kappa - \frac{16}{\kappa}.$$

Proof. (a) For $\beta \notin \frac{\pi}{2}\mathbb{Z}$, $\psi(-\beta) = -\psi(\beta)$ since sin is odd and $\tan^2$ even, and $\psi(\beta + \frac{\pi}{2}) = \psi(\beta)$ since $\sin(2\beta + \pi) = -\sin 2\beta$ and $\tan^2(\beta + \frac{\pi}{2}) = \cot^2 \beta$, so that $\log \tan^2(\beta + \frac{\pi}{2}) = -\log \tan^2 \beta$. Near 0, $|\psi(\beta)| \le 2|\beta|\,|\log \tan^2 \beta| \to 0$, so ψ is continuous at 0 with $\psi(0) = 0$, and by periodicity at every point of $\frac{\pi}{2}\mathbb{Z}$. On $(0, \frac{\pi}{4})$, $\sin 2\beta > 0$ and $0 < \tan^2 \beta < 1$, so $\psi < 0$; by oddness $\psi > 0$ on $(-\frac{\pi}{4}, 0)$. $\psi(\pm\frac{\pi}{4}) = \pm \log 1 = 0$. By periodicity these are all zeros.

(b) h_2 is smooth on $(0,1)$ with $h_2'(p) = \log \frac{1-p}{p}$. For $\beta \notin \frac{\pi}{2}\mathbb{Z}$, $0 < \cos^2 \beta < 1$ and $\Psi'(\beta) = h_2'(\cos^2 \beta) \cdot (-\sin 2\beta) = -\sin 2\beta \log \tan^2 \beta = -\psi(\beta)$. At $\beta_0 \in \frac{\pi}{2}\mathbb{Z}$, Ψ is continuous, and by the mean value theorem $(\Psi(\beta) - \Psi(\beta_0))/(\beta - \beta_0) = -\psi(\xi)$ for some ξ between β_0 and β; by continuity of ψ this tends to $-\psi(\beta_0) = 0$. Hence $\Psi \in C^1(\mathbb{R})$, $\Psi' = -\psi$. Evenness and periodicity follow from $\cos^2(-\beta) = \cos^2 \beta$ and $h_2(\cos^2(\beta + \frac{\pi}{2})) = h_2(\sin^2 \beta) = h_2(\cos^2 \beta)$.

(c) Away from $\frac{\pi}{2}\mathbb{Z}$, $(\log \tan^2 \beta)' = 2/(\sin \beta \cos \beta) = 4/\sin 2\beta$, hence

$$\psi'(\beta) = 2 \cos 2\beta \log \tan^2 \beta + 4, \qquad \psi''(\beta) = -4\psi(\beta) + 8 \cot 2\beta.$$

For $b \in (0, \frac{\pi}{4})$ let $g(b) = \cos 2b \log \tan b$, so that $\psi'(-b) = 4(g(b)+1)$. Then $g'(b) = -2 \sin 2b \log \tan b + 2 \cos 2b / \sin 2b > 0$, $g(b) \to -\infty$ as $b \to 0$ and $g(\frac{\pi}{4}) = 0$. Hence $g = -1$ at exactly one point b^*, $\psi' > 0$ on $(-\frac{\pi}{4}, -b^*)$ and $\psi' < 0$ on $(-b^*, 0)$; put $\beta_\kappa = -b^*$, $s^* = 2b^* \in (0, \frac{\pi}{2})$. Since $\log \tan b = -\operatorname{artanh}(\cos 2b)$ (both sides equal $\frac{1}{2} \log \frac{1-\cos 2b}{1+\cos 2b}$), the equation $g(b^*) = -1$ reads $y \operatorname{artanh} y = 1$ with $y = \cos s^*$, and $y \operatorname{artanh} y$ increases strictly from 0 to ∞ on $(0,1)$. Then $\psi(\beta_\kappa) = -\sin s^* \cdot 2 \log \tan b^* = 2 \sin s^* / \cos s^* = 2 \tan s^*$. By (a) the maximum of ψ over a period is attained in $[-\frac{\pi}{4}, 0]$, hence at β_κ. Finally $\psi''(\beta_\kappa) = -4\kappa + 8 \cot(-s^*) = -4\kappa - 8/\tan s^* = -4\kappa - 16/\kappa$. □

Numerically $\kappa = 1.3255$ and $\beta_\kappa \approx -0.2926$.

4.2 The variational formula

Write $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$.

Lemma 4.2. *Every unitary $U \in M_2$ can be written as $U = D_1 R_\theta D_2$ with diagonal unitaries D_1, D_2 and $\theta \in [0, \frac{\pi}{2}]$, $\cos^2 \theta = |u_{11}|^2$. Consequently $H(\Delta_2 | U \Delta_2 U^*) = H(\Delta_2 | R_\theta \Delta_2 R_\theta^T)$, and $\gamma = \operatorname{Ad} D_1$ maps optimal partitions for R_θ onto optimal partitions for U.*

Proof. The columns and the rows of U are unit vectors, so $|u_{11}|^2 + |u_{21}|^2 = 1 = |u_{11}|^2 + |u_{12}|^2$ and $|u_{12}|^2 + |u_{22}|^2 = 1$; hence $|u_{21}| = |u_{12}|$ and $|u_{22}| = |u_{11}|$. Let $\theta \in [0, \frac{\pi}{2}]$ with $\cos\theta = |u_{11}|$ and write $u_{11} = e^{ia} \cos\theta$, $u_{21} = e^{ib} \sin\theta$, $u_{12} = -e^{ic} \sin\theta$, $u_{22} = e^{id} \cos\theta$. If $0 < \theta < \frac{\pi}{2}$, orthogonality of the columns gives $(e^{i(d-b)} - e^{i(c-a)}) \cos\theta \sin\theta = 0$, i.e. $e^{i(d-b)} = e^{i(c-a)}$, and then $U = \operatorname{diag}(e^{ia}, e^{ib})\, R_\theta \operatorname{diag}(1, e^{i(c-a)})$, as one checks entry by entry. If $\theta = 0$, U is diagonal and $U = U \cdot R_0 \cdot 1$; if $\theta = \frac{\pi}{2}$, $U = \operatorname{diag}(-u_{12}, u_{21}) R_{\pi/2}$. Since $D_2 \Delta_2 D_2^* = \Delta_2$, $U \Delta_2 U^* = \gamma(R_\theta \Delta_2 R_\theta^T)$ with $\gamma = \operatorname{Ad} D_1$, which fixes Δ_2; Lemma 2.2(b) gives the rest. □

We use the Pauli matrices $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ and the Bloch parametrisation $\rho = \frac{1}{2}(1 + r \cdot \sigma)$, $r \in \mathbb{R}^3$, $|r| \le 1$, of $\mathcal{D}$ for $n = 2$; ρ is a rank-one projection if and only if $|r| = 1$. For a unit vector

$v = (\cos a, \sin a)^T$ one has $vv^* = \frac{1}{2}(1 + \sin 2a\,\sigma_1 + \cos 2a\,\sigma_3)$, so its Bloch vector is $(\sin 2a, 0, \cos 2a)$, and the Bloch vector of $(-\sin a, \cos a)^T$ is the negative of it. Let

$$z = (0,0,1), \qquad q = (\sin 2\theta, 0, \cos 2\theta), \qquad k(s) = h_2\big(\tfrac{1+s}{2}\big)\ (|s| \le 1),$$

so z and q are the Bloch vectors of e_1 and of $R_\theta e_1$, and k is even.

Lemma 4.3. *For $P = \Delta_2$, $Q = R_\theta \Delta_2 R_\theta^T$ and $\rho = \frac{1}{2}(1 + r\cdot\sigma)$, $G(\rho) = f(r) := k(r\cdot q) - k(r\cdot z)$. Moreover*

$$k(s) = \log 2 - \sum_{m\ge 1} \frac{s^{2m}}{2m(2m-1)} \qquad (|s| \le 1), \tag{8}$$

the series converging uniformly on $[-1,1]$.

Proof. $E_{\Delta_2}\rho = \operatorname{diag}(\rho)$ has the eigenvalues $\frac{1}{2}(1 \pm r_3) = \frac{1}{2}(1 \pm r\cdot z)$, so $S(E_{\Delta_2}\rho) = k(r\cdot z)$. By (4) the eigenvalues of $E_Q\rho$ are $\langle v_1, \rho v_1\rangle$ and $\langle v_2, \rho v_2\rangle = 1 - \langle v_1, \rho v_1\rangle$ with $v_1 = R_\theta e_1$, and $\langle v_1, \rho v_1\rangle = \operatorname{Tr}(\rho v_1 v_1^*) = \frac{1}{2}(1 + r\cdot q)$ by $\operatorname{Tr}\big((1 + r\cdot\sigma)(1 + q\cdot\sigma)\big) = 2(1 + r\cdot q)$. So $S(E_Q\rho) = k(r\cdot q)$.

For (8), $k(s) = \log 2 - \frac{1}{2}\big[(1+s)\log(1+s) + (1-s)\log(1-s)\big]$. For $|s| < 1$, integrating $\frac{d^2}{ds^2}(1 \pm s)\log(1\pm s) = 1/(1\pm s) = \sum_{j\ge 0}(\mp s)^j$ twice gives $(1\pm s)\log(1\pm s) = \pm s + \sum_{l\ge 2}(\mp 1)^l s^l/(l(l-1))$, and adding the two expansions leaves $2\sum_{m\ge 1} s^{2m}/(2m(2m-1))$. The terms are bounded by $1/(2m(2m-1))$ on $[-1,1]$, so the series converges uniformly there (Weierstrass), its sum is continuous, and the identity extends to $s = \pm 1$ by continuity of k. □

Theorem 4.4. *Let $U \in M_2$ be unitary, $U = D_1 R_\theta D_2$ as in Lemma 4.2, $c = |u_{11}|^2 = \cos^2\theta$, and*

$$\varphi(\alpha) = h_2\big(\cos^2(\alpha - \theta)\big) - h_2(\cos^2\alpha) = \int_{\alpha-\theta}^{\alpha} \psi(\beta)\, d\beta.$$

Then φ is C^1 and $\frac{\pi}{2}$-periodic, $\varphi'(\alpha) = \psi(\alpha) - \psi(\alpha - \theta)$, and

(a) $H(\Delta_2 | U\Delta_2 U^*) = \max_{\alpha\in\mathbb{R}} \varphi(\alpha) = G^*$;

(b) for every maximiser α^* of φ, the projections onto the orthonormal basis $D_1(\cos\alpha^*, \sin\alpha^*)^T$, $D_1(-\sin\alpha^*, \cos\alpha^*)^T$ form an optimal partition;

(c) $H(\Delta_2 | U\Delta_2 U^*)$ depends only on c, and it is invariant under $c \mapsto 1 - c$.

Proof. By Lemma 4.2 we may assume $U = R_\theta$, $D_1 = 1$, and transport optimal partitions by $\operatorname{Ad} D_1$ at the end. The integral form is $\varphi(\alpha) = \Psi(\alpha - \theta) - \Psi(\alpha) = \int_{\alpha-\theta}^{\alpha} \psi$ by Lemma 4.1(b), which also gives the regularity, periodicity and derivative of φ.

Step 1: $H = G^* = \max_{|r|\le 1} f(r)$. This is Lemma 3.2 together with Lemma 4.3; in the Bloch picture, $f(-r) = f(r)$ because k is even, and if r^* maximises f, the partition $\{\rho_+, \rho_-\}$ with $\rho_\pm = \frac{1}{2}(1 \pm r^*\cdot\sigma)$ is optimal.

Step 2: reduction to the unit circle in the plane of z and q. If $\theta \in \{0, \frac{\pi}{2}\}$, then $q = \pm z$ and $f \equiv 0$ since k is even; also $\varphi \equiv 0$, because $h_2(\cos^2(\alpha - \frac{\pi}{2})) = h_2(\sin^2\alpha) = h_2(\cos^2\alpha)$. So $H = 0 = \max\varphi$ and every basis is optimal. Let $0 < \theta < \frac{\pi}{2}$. Then z and q span the plane $\{r_2 = 0\}$, and $f(r)$ depends only on the orthogonal projection $a = (r_1, 0, r_3)$ of r onto it, $|a| \le |r| \le 1$. Write $a = \varrho(\sin\omega, 0, \cos\omega)$ with $0 \le \varrho \le 1$ and the polar angle ω measured from z; then $a\cdot z = \varrho\cos\omega$ and $a\cdot q = \varrho\cos(\omega - 2\theta)$, and by (8)

$$f(a) = \sum_{m\ge 1} \frac{\varrho^{2m}}{2m(2m-1)}\big(\cos^{2m}\omega - \cos^{2m}(\omega - 2\theta)\big).$$

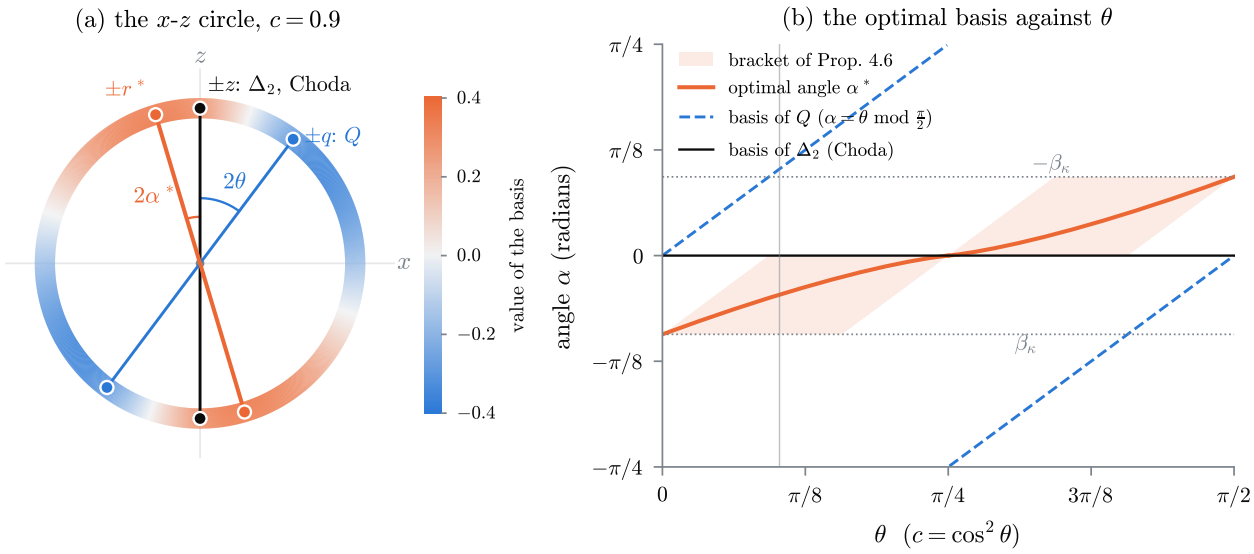


Figure 1: (a) The great circle $r_2 = 0$ of the Bloch ball for $c = 0.9$ ($\theta \approx 0.3218$). An orthonormal basis of $\mathbb{C}^2$ with real coefficients is a pair of antipodal points $\pm r$ at the polar angle $\omega = 2\alpha$ from z, and the ring is coloured by its value $\varphi(\alpha)$ (Theorem 4.4). Black: the basis of Δ_2, i.e. Choda's partition (value $h(U)$); blue: the basis of $Q = U\Delta_2 U^*$ (value $-h(U)$); orange: the optimal basis $\pm r^*$, at the angle $2\alpha^* \approx -0.290$ on the other side of z than q. (b) The optimal angle $\alpha^* \in (-\frac{\pi}{4}, \frac{\pi}{4}]$ against θ, with the bracket of Proposition 4.6 shaded; $\alpha^* \to \beta_\kappa$ as $\theta \to 0$, $\alpha^* = 0$ at $\theta = \frac{\pi}{4}$, and $\alpha^*(\theta) = -\alpha^*(\frac{\pi}{2} - \theta)$. The thin vertical line marks $c = 0.9$.

If $|\cos\omega| \geq |\cos(\omega - 2\theta)|$, all terms are non-negative and non-decreasing in ϱ, so $f(a) \leq f(\sin\omega, 0, \cos\omega)$. Otherwise all terms are non-positive and $f(a) \leq 0 = f(\sin\theta, 0, \cos\theta)$, the value at $\omega = \theta$, where $\cos\omega = \cos(\omega - 2\theta)$. Hence the maximum of f over the ball equals its maximum over the unit circle of the plane,

$$\max_{|r|\leq 1} f = \max_{\omega\in\mathbb{R}} \big[k(\cos(\omega - 2\theta)) - k(\cos\omega)\big],$$

and it is attained at a unit vector $r^* = (\sin\omega^\circ, 0, \cos\omega^\circ)$.

Step 3: the angle. Since $k(\cos 2\gamma) = h_2(\frac{1+\cos 2\gamma}{2}) = h_2(\cos^2\gamma)$, the substitution $\omega = 2\alpha$ turns the right-hand side into $\max_\alpha \varphi(\alpha)$. This proves (a). For a maximiser α^* of φ, $r^* = (\sin 2\alpha^*, 0, \cos 2\alpha^*)$ maximises f, and ρ_+, ρ_- are the projections onto $(\cos\alpha^*, \sin\alpha^*)^T$ and $(-\sin\alpha^*, \cos\alpha^*)^T$; with Step 1 and Lemma 4.2 this proves (b).

(c) φ depends only on θ, i.e. on c. Let $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $UJ\Delta_2(UJ)^* = U\Delta_2 U^*$ and $|(UJ)_{11}|^2 = |u_{12}|^2 = 1 - c$. □

Remark 4.5. (i) At $\alpha = 0$, $\varphi(0) = h_2(c) = h(U)$, the value of Choda's partition e_{11}, e_{22}. At $\alpha = \theta$ (the basis of Q) $\varphi(\theta) = -h(U)$. (ii) In the Bloch picture (Figure 1), z and q enclose the angle 2θ, and the optimal Bloch vector r^* has the angle $2\alpha^*$ from z. By Proposition 4.6 below, for $\frac{1}{2} < c < 1$ it lies on the other side of z than q. (iii) Let $0 < c < 1$. Since $H = G^*$, the optimal partitions are exactly the families $x_j = 2t_j\rho_j$ with ρ_j maximising G and $\sum_j t_j\rho_j = \frac{1}{2}$ (Proposition 3.1). Every maximiser r of f has $|r| = 1$ and $r_2 = 0$: as $f(r) = H \geq h(U) > 0$, in Step 2 of the proof we are in the case $|\cos\omega| > |\cos(\omega - 2\theta)|$, where every term of the series is strictly increasing in ϱ, and $r_2 \neq 0$ or $|r| < 1$ would give $\varrho < 1$. Since the maximiser of φ is unique modulo $\frac{\pi}{2}$ (Proposition 4.6), the maximisers of G are the two pure states $\rho_\pm$, and the optimal partition is unique up to splitting $\rho_\pm$ into several multiples (Lemma 2.2(a)).

4.3 The optimal angle

Proposition 4.6. *Let $\frac{1}{2} < c < 1$, i.e. $0 < \theta < \frac{\pi}{4}$, and let β_κ be as in Lemma 4.1(c). Then φ has a unique maximiser α^* in $(-\frac{\pi}{4}, \frac{\pi}{4}]$; it satisfies*

$$\max\bigl(\beta_\kappa, \theta - \tfrac{\pi}{4}\bigr) < \alpha^* < \min\bigl(\beta_\kappa + \theta, 0\bigr),$$

and it is the unique solution of $\psi(\alpha) = \psi(\alpha - \theta)$ in this interval. In particular $\alpha^* < 0$, and $\alpha^* \to \beta_\kappa$ as $\theta \to 0$. For $0 < c < \frac{1}{2}$ the maximiser is $\alpha^*(c) = -\alpha^*(1-c)$. For $c = \frac{1}{2}$ the maximiser is $\alpha^* = 0$ (modulo $\frac{\pi}{2}$) and $H = \log 2$.

Proof. Let $I_\ell = (-\frac{\pi}{4} + \ell\frac{\pi}{2}, \ell\frac{\pi}{2})$ and $N_\ell = (\ell\frac{\pi}{2}, \frac{\pi}{4} + \ell\frac{\pi}{2})$, $\ell \in \mathbb{Z}$; by Lemma 4.1(a), $\psi > 0$ on the I_ℓ and $\psi < 0$ on the N_ℓ, all of length $\frac{\pi}{4}$. Let $0 < \theta \le \frac{\pi}{4}$ and let $W = [\alpha - \theta, \alpha]$ be a window. If W met two of the I_ℓ, it would contain the closure of some N_ℓ together with points on both sides of it, so its length would exceed $\frac{\pi}{4}$; hence W meets at most one I_ℓ, in an interval J of length at most θ. Choose a window $W' \subset \overline{I_\ell}$ of length θ containing J. Since $\psi \ge 0$ on $\overline{I_\ell}$ and $\psi \le 0$ on $W \setminus J$,

$$\int_W \psi \le \int_J \psi \le \int_{W'} \psi,$$

and the first inequality is strict unless $W \setminus J$ is a null set, i.e. unless $W \subset \overline{I_\ell}$. Hence every maximising window lies in some $\overline{I_\ell}$; modulo $\frac{\pi}{2}$, $W \subset [-\frac{\pi}{4}, 0]$, i.e. $\alpha \in [\theta - \frac{\pi}{4}, 0]$.

$c = \frac{1}{2}$. For $\theta = \frac{\pi}{4}$ the only such window is $[-\frac{\pi}{4}, 0]$, so $\alpha^* = 0$ and $H = \Psi(-\frac{\pi}{4}) - \Psi(0) = h_2(\frac{1}{2}) - h_2(1) = \log 2$.

$\frac{1}{2} < c < 1$. Now $0 < \theta < \frac{\pi}{4}$. At the end points, $\varphi'(0) = \psi(0) - \psi(-\theta) = -\psi(-\theta) < 0$ and $\varphi'(\theta - \frac{\pi}{4}) = \psi(\theta - \frac{\pi}{4}) - \psi(-\frac{\pi}{4}) = \psi(\theta - \frac{\pi}{4}) > 0$, so neither is a maximiser of the C^1 function φ on $\mathbb{R}$, and every maximiser α^* (modulo $\frac{\pi}{2}$) lies in $(\theta - \frac{\pi}{4}, 0)$ and satisfies $\varphi'(\alpha^*) = 0$, i.e. $\psi(\alpha^*) = \psi(\alpha^* - \theta)$ with $\alpha^* - \theta < \alpha^*$ both in $(-\frac{\pi}{4}, 0)$. As ψ is strictly monotone on $[-\frac{\pi}{4}, \beta_\kappa]$ and on $[\beta_\kappa, 0]$ (Lemma 4.1(c)), two distinct points with equal values lie on different sides of β_κ: $\alpha^* - \theta < \beta_\kappa < \alpha^*$. On the interval $L = (\max(\beta_\kappa, \theta - \frac{\pi}{4}), \min(\beta_\kappa + \theta, 0))$ we have $\alpha \in (\beta_\kappa, 0)$ and $\alpha - \theta \in (-\frac{\pi}{4}, \beta_\kappa)$, so $\psi(\alpha)$ is strictly decreasing and $\psi(\alpha - \theta)$ strictly increasing in α; hence φ' is strictly decreasing on L and has at most one zero there. A maximiser exists by continuity and periodicity, so it is unique modulo $\frac{\pi}{2}$. As $\theta \to 0$, $L \subset (\beta_\kappa, \beta_\kappa + \theta)$ shrinks to β_κ.

$0 < c < \frac{1}{2}$. With $\theta' = \frac{\pi}{2} - \theta$ and Lemma 4.1(b), $\varphi_\theta(\alpha) = \Psi(\alpha - \frac{\pi}{2} + \theta') - \Psi(\alpha) = \Psi(-\alpha - \theta') - \Psi(-\alpha) = \varphi_{\theta'}(-\alpha)$, where φ_θ denotes φ for the angle θ; and $\cos^2\theta' = 1 - c$. □

The proof is the sliding-window picture of Figure 2: a window of length $\theta \le \frac{\pi}{4}$ that captures the most area under ψ must lie in a single positive hump of ψ, and within the hump it is optimal when the values of ψ at its two ends agree. As $\theta \to 0$ the optimal basis therefore does not approach the basis of Δ_2 (angle 0) or of Q (angle θ), but the basis at the fixed angle β_κ where ψ is largest (Figure 1(b)).

5 Bounds and asymptotics for $n = 2$

Theorem 5.1. *Let $U \in M_2$ be unitary, $c = |u_{11}|^2 = \cos^2\theta$, $\theta \in [0, \frac{\pi}{2}]$, and $\kappa = \max\psi$ as in Lemma 4.1(c). Then*

(a) $h(U) = h_2(c) \le H(\Delta_2 | U\Delta_2 U^*) \le \kappa \min\bigl(\theta, \frac{\pi}{2} - \theta\bigr)$;

(b) as $\theta \to 0$,

$$H(\Delta_2 | U\Delta_2 U^*) = \kappa\theta - \frac{\kappa + 4/\kappa}{6}\theta^3 + O(\theta^4), \qquad h(U) = 2\theta^2 \log\frac{1}{\theta} + \theta^2 + O\Bigl(\theta^4 \log\frac{1}{\theta}\Bigr);$$

in particular $H/h(U) \to \infty$ as $c \to 1$, and likewise as $c \to 0$.

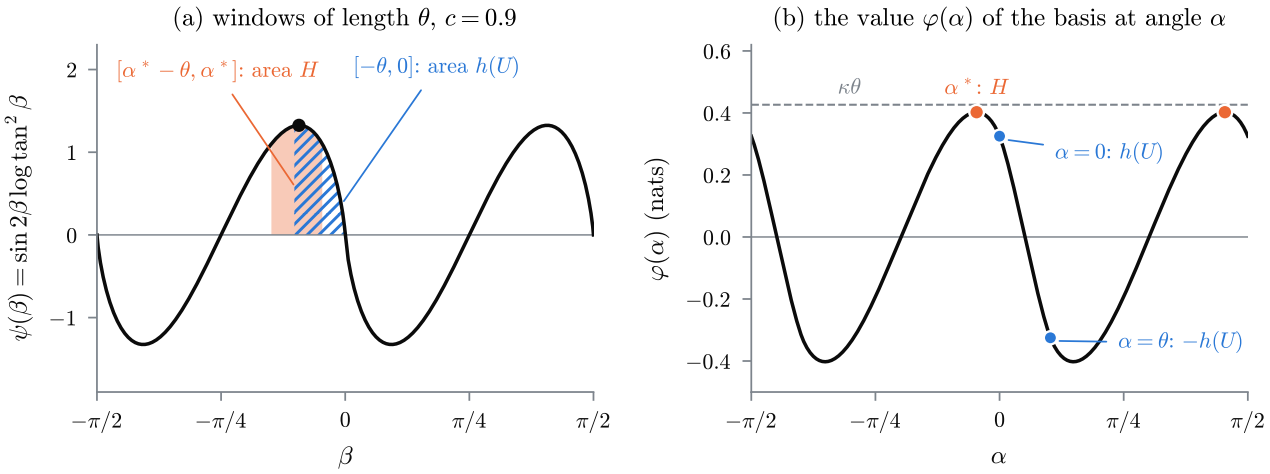


Figure 2: (a) The function ψ for $c = 0.9$. By Theorem 4.4 the value of the basis at angle α is the area under ψ over the window $[\alpha - \theta, \alpha]$. Choda's partition corresponds to the window $[-\theta, 0]$ (hatched, area $h(U)$); the optimal basis to the window of the same length with the largest area, which straddles the maximum $\kappa = \psi(\beta_\kappa)$ (dot) and whose end points have equal values of ψ (shaded, area H). (b) The value $\varphi(\alpha)$ of the basis at angle α, with the three bases of Figure 1 and the upper bound $\kappa\theta$ of Theorem 5.1.

Proof. (a) The lower bound is $\varphi(0) = h_2(c)$. For every α, $\varphi(\alpha) = \int_{\alpha-\theta}^{\alpha} \psi \le \theta \max \psi = \kappa\theta$, so $H \le \kappa\theta$ by Theorem 4.4(a). Applying this to $1 - c$, whose angle is $\frac{\pi}{2} - \theta$, and using Theorem 4.4(c) gives $H \le \kappa(\frac{\pi}{2} - \theta)$.

(b) Fix $\delta > 0$ with $[\beta_\kappa - \delta, \beta_\kappa + \delta] \subset (-\frac{\pi}{4}, 0)$; ψ is smooth there, and by Taylor's theorem there is C with $|\psi(\beta) - \kappa - \frac{1}{2}\psi''(\beta_\kappa)(\beta - \beta_\kappa)^2| \le C|\beta - \beta_\kappa|^3$ for $|\beta - \beta_\kappa| \le \delta$. Let $\theta < \min(\delta, \frac{\pi}{4})$. By Proposition 4.6 the maximising window $W = [\alpha^* - \theta, \alpha^*]$ contains β_κ, so $W \subset [\beta_\kappa - \delta, \beta_\kappa + \delta]$. Since $\int_W (\beta - \beta_\kappa)^2 d\beta \ge \theta^3/12$ for every window of length θ (with equality for the window centred at β_κ), $\psi''(\beta_\kappa) < 0$ and $\int_W |\beta - \beta_\kappa|^3 d\beta \le \theta^4$,

$$\kappa\theta + \frac{\psi''(\beta_\kappa)}{24}\theta^3 - C\theta^4 \;\le\; \varphi\big(\beta_\kappa + \tfrac{\theta}{2}\big) \;\le\; H \;=\; \int_W \psi \;\le\; \kappa\theta + \frac{\psi''(\beta_\kappa)}{24}\theta^3 + C\theta^4,$$

and $\psi''(\beta_\kappa)/24 = -(\kappa + 4/\kappa)/6$ by Lemma 4.1(c).

For Choda's value put $s = \sin^2\theta = 1 - c$, so $h(U) = \eta(s) + \eta(1-s)$. Since $-\log(1-s) = s + O(s^2)$, $\eta(1-s) = s + O(s^2) = \theta^2 + O(\theta^4)$. Since $s = \theta^2(1 + O(\theta^2))$ and $\log s = 2\log\theta + O(\theta^2)$, $\eta(s) = -s\log s = 2\theta^2 \log\frac{1}{\theta} + O(\theta^4 \log\frac{1}{\theta})$. Adding gives the expansion of $h(U)$, and $H/h(U) \sim \kappa/(2\theta\log\frac{1}{\theta}) \to \infty$. The case $c \to 0$ follows from Theorem 4.4(c) and the symmetry $h_2(c) = h_2(1-c)$. □

Since $\theta = \arccos\sqrt{c} \sim \sqrt{1-c}$, near $c = 1$ the entropy behaves like $\kappa\sqrt{1-c}$, much larger than $h(U) \sim (1-c)\log\frac{1}{1-c}$. The bound $\log 2$ of Proposition 2.3(c) is better than $\kappa\min(\theta, \frac{\pi}{2} - \theta)$ near $c = \frac{1}{2}$. Figure 3 compares H with $h(U)$ over the whole range and checks the two expansions of Theorem 5.1(b), and Table 1 lists values.

Corollary 5.2 (Petz–Szántó–Weiner). *$H(\Delta_2|U\Delta_2U^*) > h(U)$ unless $c \in \{0, \frac{1}{2}, 1\}$, where $H = h(U)$.*

Proof. This is [18, Appendix]; in terms of Theorem 4.4 it reads as follows. $h(U) = h_2(c) = \varphi(0)$ and $\varphi'(0) = \psi(0) - \psi(-\theta) = \psi(\theta)$, which by Lemma 4.1(a) vanishes exactly for $\theta \in \{0, \frac{\pi}{4}, \frac{\pi}{2}\}$. Otherwise 0 is not a maximiser of φ, so $H > \varphi(0)$. For $c \in \{0, 1\}$, $H = 0 = h_2(c)$, and for $c = \frac{1}{2}$, $H = \log 2 = h_2(\frac{1}{2})$ (Proposition 4.6). □

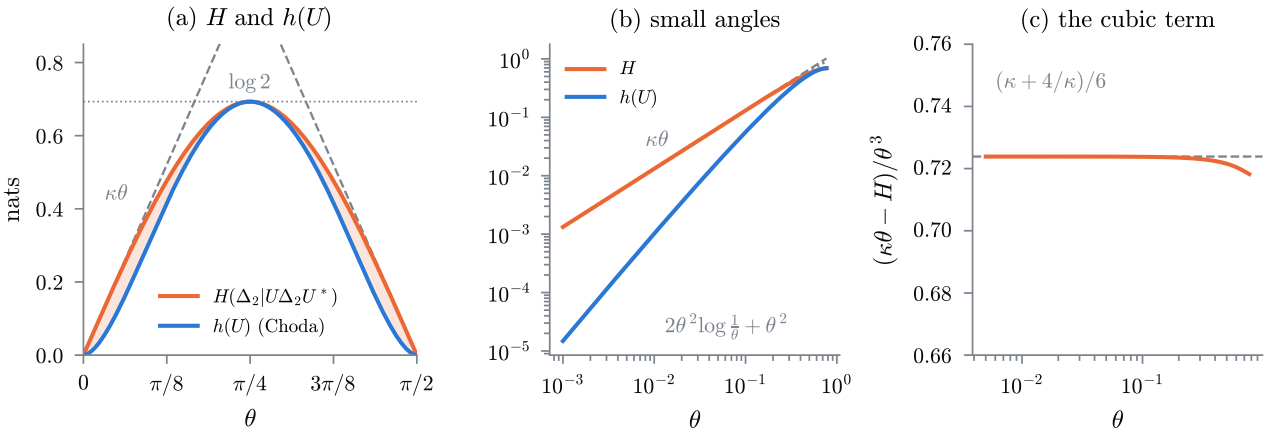


Figure 3: (a) $H(\Delta_2|U\Delta_2U^*)$ (Theorem 4.4) and Choda's value $h(U)$ against θ; the shaded gap $H-h(U)$ vanishes exactly at $\theta\in\{0,\frac{\pi}{4},\frac{\pi}{2}\}$ (Corollary 5.2). Dashed: the bound $\kappa\min(\theta,\frac{\pi}{2}-\theta)$; dotted: $\log 2$. (b) The same on logarithmic axes for small θ, with the leading terms $\kappa\theta$ and $2\theta^2\log\frac{1}{\theta}+\theta^2$ of Theorem 5.1(b). (c) $(\kappa\theta-H)/\theta^3$, which tends to $(\kappa+4/\kappa)/6\approx 0.724$. All curves are computed from the formulas of Section 4.

c	$h(U)=h_2(c)$	H, Theorem 4.4	H, numerical search	α^* (radians)	$\kappa\theta$
0.99	0.056002	0.132043	0.132043	-0.2438	0.132771
0.95	0.198515	0.290618	0.290618	-0.1864	0.298915
0.9	0.325083	0.402396	0.402396	-0.1452	0.426476
0.8	0.500402	0.542607	0.542607	-0.0899	0.614559
0.7	0.610864	0.627949	0.627949	-0.0504	0.768305
0.6	0.673012	0.676658	0.676658	-0.0205	0.907586
0.5	0.693147	0.693147	0.693147	+0.0000	1.041035

Table 1: $H(\Delta_2|U\Delta_2U^*)$ for $c=|u_{11}|^2$: Choda's value, the variational formula of Theorem 4.4 with its optimal angle $\alpha^*\in(-\frac{\pi}{4},\frac{\pi}{4}]$, the maximum over general partitions with four elements found by the numerical search of Appendix A for $U=D_1R_\theta D_2$ with random phases, and the upper bound $\kappa\theta$ of Theorem 5.1.

Remark 5.3 (Spin model subfactors). Bakshi and Guin use the inequality $H(u\Delta_2u^*|v\Delta_2v^*)\le H(R_u|R_v)$ for the spin model subfactors R_u,R_v of two 2×2 complex Hadamard matrices u,v [1, proof of Theorem 6.22], together with Choda's bound. By Lemma 2.2(b) with $\gamma=\mathrm{Ad}\,u^*$, $H(u\Delta_2u^*|v\Delta_2v^*)=H(\Delta_2|U\Delta_2U^*)$ with $U=u^*v$, so Theorem 4.4 with $c=|(u^*v)_{11}|^2$ raises their lower bound $h_2(c)$ for $H(R_u|R_v)$ to $\max_\alpha\varphi(\alpha)$, e.g. from 0.3251 to 0.4024 at $c=0.9$ (Table 1).

6 Masas of M_n for $n\ge 3$

6.1 Choda's partition is generically not optimal

For $n=2$, Corollary 5.2 was obtained by differentiating the value of a basis at Choda's partition. The same first-order argument works in every dimension. For a unitary $U\in M_n$ let

$$A_U=\big(u_{jk}\log|u_{jk}|^2\big)_{j,k=1}^n\quad(\text{with }0\log 0=0),\qquad M_U=A_UU^*.$$

The map $U\mapsto M_U$ is continuous, since $t\mapsto t\log|t|^2$ is continuous on $\mathbb{C}$.

Proposition 6.1. *Let $U \in M_n$ be unitary, $X \in M_n$ skew-Hermitian, $W_\varepsilon = \exp(\varepsilon X)$, and let $V(\varepsilon) = h(U^*W_\varepsilon) - h(W_\varepsilon)$ be the value (6) of the basis W_ε. Then $V(0) = h(U)$ and*

$$\lim_{\varepsilon \downarrow 0} \frac{V(\varepsilon) - V(0)}{\varepsilon} = -\frac{2}{n} \operatorname{Re} \operatorname{Tr}(M_U X).$$

Consequently, if M_U is not Hermitian, then $H(\Delta_n | U \Delta_n U^*) > h(U)$; an improving direction is $X = \frac{1}{2}(M_U - M_U^*)$, for which the derivative equals $\frac{2}{n}\|X\|_2^2 > 0$. The set of unitaries with $M_U \neq M_U^*$ is open, and for every $n \geq 2$ it contains $R_\theta \oplus 1_{n-2}$ for $\theta \notin \frac{\pi}{4}\mathbb{Z}$; in particular it has positive Haar measure.

Proof. $V(0) = h(U^*) - h(1) = h(U)$, since $|(U^*)_{ij}| = |u_{ji}|$ and $h(1) = 0$.

The term $h(W_\varepsilon)$ is $o(\varepsilon)$. $W_\varepsilon = 1 + \varepsilon X + O(\varepsilon^2)$, so $|(W_\varepsilon)_{ij}|^2 \leq C\varepsilon^2$ for $i \neq j$, and $1 - |(W_\varepsilon)_{ii}|^2 = \sum_{j \neq i} |(W_\varepsilon)_{ij}|^2 \leq C\varepsilon^2$ because the rows are unit vectors. As $\eta(1-\delta) \leq \delta$ for $\delta \in [0,1]$ and η is increasing on $[0, e^{-1}]$, $h(W_\varepsilon) \leq C'(\varepsilon^2 + \varepsilon^2 \log \frac{1}{\varepsilon}) = o(\varepsilon)$ for small ε, and $h \geq 0$.

The term $h(U^*W_\varepsilon)$. Let $b_{ij}(\varepsilon) = (U^*W_\varepsilon)_{ij} = \overline{u_{ji}} + \varepsilon (U^*X)_{ij} + O(\varepsilon^2)$. If $u_{ji} = 0$, then $|b_{ij}|^2 = O(\varepsilon^2)$ and $\eta(|b_{ij}|^2) = o(\varepsilon)$ as above. If $u_{ji} \neq 0$, $\varepsilon \mapsto \eta(|b_{ij}(\varepsilon)|^2)$ is differentiable at 0 with derivative $\eta'(|u_{ji}|^2) \cdot 2 \operatorname{Re}\big(u_{ji}(U^*X)_{ij}\big) = -2\big(1 + \log|u_{ji}|^2\big) \operatorname{Re}\big(u_{ji}(U^*X)_{ij}\big)$. Summing over i, j (the terms with $u_{ji} = 0$ may be included, as they vanish),

$$\begin{aligned} \frac{d}{d\varepsilon}\Big|_{0^+} h(U^*W_\varepsilon) &= -\frac{2}{n} \operatorname{Re} \sum_{i,j} u_{ji}(U^*X)_{ij} - \frac{2}{n} \operatorname{Re} \sum_{i,j} (A_U)_{ji}(U^*X)_{ij} \\ &= -\frac{2}{n} \operatorname{Re} \operatorname{Tr}(X) - \frac{2}{n} \operatorname{Re} \operatorname{Tr}(A_U U^* X). \end{aligned}$$

$\operatorname{Tr} X$ is purely imaginary, so the first term vanishes, and the second is $-\frac{2}{n}\operatorname{Re}\operatorname{Tr}(M_U X)$.

The improving direction. Write $M_U = M_h + M_a$ with $M_h = \frac{1}{2}(M_U + M_U^*)$ and $M_a = \frac{1}{2}(M_U - M_U^*)$. For skew-Hermitian X, $\operatorname{Tr}(M_h X)$ is purely imaginary and $\operatorname{Tr}(M_a X)$ is real. With $X = M_a$, $\operatorname{Re}\operatorname{Tr}(M_U X) = \operatorname{Tr}(M_a^2) = -\operatorname{Tr}(M_a^* M_a) = -\|M_a\|_2^2$, so the derivative is $\frac{2}{n}\|M_a\|_2^2$, positive if M_U is not Hermitian. Then $V(\varepsilon) > h(U)$ for small $\varepsilon > 0$, and $V(\varepsilon) \leq H$.

The example. The set is open because $U \mapsto M_U$ is continuous. For $U = R_\theta \oplus 1_{n-2}$ write $c_\theta = \cos\theta$, $s_\theta = \sin\theta$. The diagonal entries 1 and the zero entries contribute nothing to A_U, and $M_U = M_2 \oplus 0$ with

$$M_2 = \begin{pmatrix} c_\theta \log c_\theta^2 & -s_\theta \log s_\theta^2 \\ s_\theta \log s_\theta^2 & c_\theta \log c_\theta^2 \end{pmatrix} \begin{pmatrix} c_\theta & s_\theta \\ -s_\theta & c_\theta \end{pmatrix} = \begin{pmatrix} c_\theta^2 \log c_\theta^2 + s_\theta^2 \log s_\theta^2 & c_\theta s_\theta \log \cot^2\theta \\ -c_\theta s_\theta \log \cot^2\theta & c_\theta^2 \log c_\theta^2 + s_\theta^2 \log s_\theta^2 \end{pmatrix},$$

which is real, and Hermitian if and only if $c_\theta s_\theta \log \cot^2\theta = 0$, i.e. $\theta \in \frac{\pi}{4}\mathbb{Z}$. A non-empty open subset of $U(n)$ has positive Haar measure. □

For $n = 2$ and $U = R_\theta$ the criterion is exactly Corollary 5.2, so Proposition 6.1 is the n-dimensional form of the argument of Petz, Szántó and Weiner. We expect M_U to be non-Hermitian for Haar-almost every U, and the computations below find $H > h(U)$ for every sampled unitary, but we have not proved the almost-everywhere statement.

6.2 Numerical findings

For $n \geq 3$ we know no analogue of Lemma 3.2, and we turned to computation. For Haar-random unitaries U (300 for $n = 3$, 150 for $n = 4$, 24 for $n = 5$) we maximised the partition value $\sum_j F(x_j)$ over partitions with n^2 elements, which suffice by Proposition 2.3(b), and separately the value $h(U^*W) - h(W)$ over orthonormal bases W, and G over states, each by multistart local optimisation; whenever a partition appeared to beat the bases, we repeated the basis search with many more starts, including starts built from the directions of the best partition. The method is described in Appendix A.

n	U	starts	partition $>$ basis	largest gap	agreement otherwise	$H - h(U)$: min, median, max			$G^* > H$
3	300	8/8	22	7×10^{-3}	8×10^{-12}	0.0002	0.0322	0.1333	300
4	150	8/8	6	3×10^{-4}	1×10^{-11}	0.0047	0.0327	0.1762	150
5	24	16/16	2	5×10^{-4}	7×10^{-13}	0.0122	0.0323	0.0849	24

Table 2: Haar-random $U \in U(n)$: the number of unitaries, the starts per unitary for general partitions (n^2 elements) and for bases; the number of unitaries for which a partition beats the best basis found (after the extended basis search) and the largest such gap (both values recomputed with 50 digits); the largest difference between the best partition and the best basis for the others; the excess of the best value found, H, over Choda's $h(U)$; and the number of unitaries with $G^* > H + 10^{-6}$.

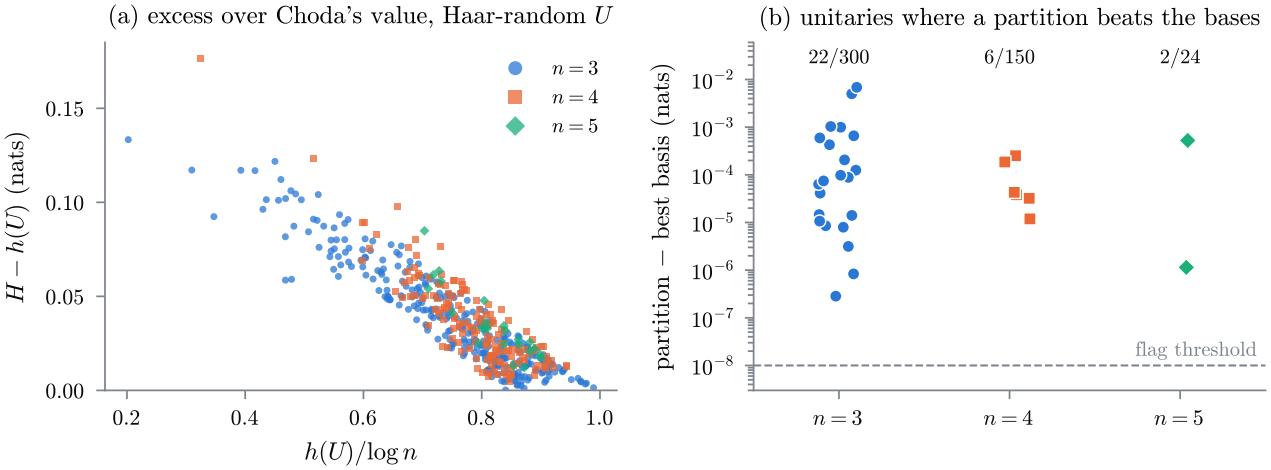


Figure 4: Haar-random unitaries, $n = 3, 4, 5$. (a) The excess $H - h(U)$ of the best value found over Choda's value, against the normalised Choda value $h(U)/\log n$; the excess vanishes as $h(U) \to \log n$ (complementary masas). (b) For the unitaries at which the best partition found beats the best basis found by more than 1×10^{-8}: the difference of the two values, both recomputed with 50 digits at explicit points; the fractions give the number of such unitaries.

Status of the findings. Apart from Proposition 6.1, every statement about $n \geq 3$ in this section is a numerical observation and is labelled as such once and for all here. Each value reported for a partition, a basis or a state is the value of the functional at an explicit point; where a comparison depends on it, the point was recomputed with 50 significant digits from an exactly normalised partition, so these values are rigorous lower bounds for H (or G^*) up to rounding errors far below the reported digits. What is not certified is that any of the maxima is global: the best basis found need not be the best basis.

Table 2 and Figure 4 summarise the results; we describe them in order of increasing departure from the case $n = 2$.

Bases are usually optimal. For most unitaries the best general partition found is, up to the merging of parallel elements (Lemma 2.2(a)), an orthonormal basis: the best partition and the best basis agree to 8×10^{-12}, 1×10^{-11} and 7×10^{-13} for $n = 3, 4, 5$. For $n = 2$ this is Theorem 4.4.

But not always. For 22 of the 300 unitaries with $n = 3$, 6 of the 150 with $n = 4$ and 2 of the 24 with $n = 5$, a partition beats the best basis found by more than 1×10^{-8}, and none of these gaps was closed by the extended basis search. The gaps range from 3×10^{-7} to 6.9×10^{-3} for $n = 3$, from 1×10^{-5} to 2.5×10^{-4} for $n = 4$ and from 1×10^{-6} to 5.3×10^{-4} for $n = 5$ (Figure 4(b)). For the

unitary U' with the largest gap the values at the two explicit points are

n	partition (a lower bound for $H(\Delta_n \mid U'\Delta_n U'^*)$)	best basis found
3	0.5727784466	0.5659197506
4	1.0949192541	1.0946676581
5	1.1963669266	1.1958404880

The structure of the better partitions. In all these cases every element of the best partition has rank one, and the elements point in 4, 5 and 6 distinct directions for $n = 3, 4, 5$, that is, in $n+1$ directions; by Lemma 2.2(a) the partition is, up to the merging tolerance, a partition into $n+1$ rank-one elements. They are not small perturbations of a basis: two of the directions are far from orthogonal, with the largest overlap $|\langle d, d'\rangle|$ equal to 0.897, 0.980 and 0.991 for the unitaries with the largest gap.

Choda's partition is never optimal. In every sample the best value found exceeds $h(U)$, by at least 0.0002 ($n = 3$; Table 2), as predicted by Proposition 6.1 when M_U is not Hermitian. The excess is largest when $h(U)$ is small and vanishes as $h(U) \to \log n$, in accordance with the characterisation of complementary pairs (Figure 4(a)).

The envelope bound is not attained. For every sampled unitary the computed G^* exceeds the computed H, by at least 0.004, 0.026 and 0.076 for $n = 3, 4, 5$. Unlike for $n = 2$, the optimal ensembles do not appear to consist of maximisers of G (Remark 3.3(i)).

Observation 6.2. For $n = 3, 4, 5$ and a small but non-negligible fraction of Haar-random unitaries U, the supremum (1) for $H(\Delta_n \mid U\Delta_n U^*)$ is not attained at an orthonormal basis: partitions with $n+1$ rank-one elements are better than every basis found by an extensive multistart search.

Conjecture 6.3. Let $n \geq 3$.

(a) There are unitaries $U \in U(n)$ (in fact, a set of positive Haar measure) with $H(\Delta_n \mid U\Delta_n U^*) > \max_{W \in U(n)} [h(U^*W) - h(W)]$.

(b) For Haar-almost every U, $h(U) < H(\Delta_n \mid U\Delta_n U^*) < G^*$.

6.3 What would make the observations theorems

Conjecture 6.3(a) would follow from a certified upper bound for the best basis at a single unitary. Indeed, the value of a fixed partition and the function $B(U) = \max_W [h(U^*W) - h(W)]$ are both continuous in U (the latter as the maximum of a jointly continuous function over the compact group $U(n)$), so a strict inequality between them at one unitary persists on a neighbourhood, which has positive Haar measure. Since $h(U^*W) - h(W)$ is invariant under $W \mapsto WD\Pi$ for diagonal unitaries D and permutation matrices Π, the maximisation runs over the flag manifold $U(n)/\mathbb{T}^n$, of real dimension $n(n-1)$, i.e. six for $n = 3$; for the stored 3×3 unitary with the largest gap the margin to be certified is about 7×10^{-3}, which seems within reach of interval branch-and-bound. The lower half of Conjecture 6.3(b) is Proposition 6.1 together with an almost-everywhere statement for the condition $M_U \neq M_U^*$. For the upper half, the computed G^* is a certified lower bound for G^*, since it is attained at an explicit state, and what is needed is a certified upper bound for H; the dual bound (7) reduces this to exhibiting an affine function a with $a(\frac{1}{n}) < G^*$ and verifying $a \geq G$ on the (n^2-1)-dimensional state space, again a single global inequality per unitary.

Remark 6.4 (Non-abelian subalgebras). The same computation applies to any pair of subalgebras of M_n. For $P = M_2 \oplus \mathbb{C} \subset M_3$ and $P = M_2 \otimes 1 \subset M_4$ against uPu^* for two Haar-random u each, the maxima over partitions with n, $2n$ and n^2 elements agree to 1×10^{-13} and 3×10^{-6}; we obtain

$H = 0.6319, 0.5142$ and $H = 0.6795, 0.6923$, against $G^* = 0.6931, 0.6931$ and $G^* = 0.6931, 0.6931$. For $M_2 \otimes 1 \subset M_4$, $E_P\rho = \mathrm{Tr}_2\,\rho \otimes \frac{1}{2}$ with the partial trace over the second factor, so $S(E_P\rho) = S(\mathrm{Tr}_2\,\rho) + \log 2 \geq \log 2$, and similarly $S(E_Q\rho) \leq 2\log 2$; hence $G^* \leq \log 2$, a bound that the computed G^* reaches while the computed H stays below it.

7 Open problems

(1) *A certificate for Observation 6.2.* Prove Conjecture 6.3(a) for one explicit unitary, e.g. the stored 3×3 unitary with the largest gap, by a rigorous upper bound for $\max_W [h(U^*W) - h(W)]$ (Section 6.3).

(2) *Explicit formulas for $n \geq 3$.* Is there a formula for $H(\Delta_n | U\Delta_n U^*)$ for special families, e.g. $U = R_\theta \oplus 1_{n-2}$, unitaries close to permutation matrices, or U close to a complementary one? Is H piecewise analytic in U, with the switch between bases and partitions with $n+1$ elements as a transition?

(3) *Small angles for $n \geq 3$.* For $n = 2$, H is of first order in the angle θ, while $h(U)$ is of order $\theta^2 \log \frac{1}{\theta}$ (Theorem 5.1). Is $H(\Delta_n | U\Delta_n U^*)$ of first order in the distance of U from the monomial unitaries for every n, and with which constant?

(4) *The number of elements.* Is the supremum for two masas always attained by a partition with at most $n+1$ elements of rank one? Where partitions beat bases, the best partitions found have rank-one elements in $n+1$ directions, and otherwise a basis is as good as the best partition found; Proposition 2.3(b) only guarantees n^2 elements.

(5) *The gap to G^*.* Characterise the pairs P, Q with $H(P|Q) = G^*$ (Proposition 3.1); for masas, is equality restricted to $n = 2$ and to complementary or commuting pairs?

(6) *Almost everywhere.* Is M_U non-Hermitian for Haar-almost every $U \in U(n)$, so that $H(\Delta_n | U\Delta_n U^*) > h(U)$ almost everywhere (Proposition 6.1)?

(7) *Spin model subfactors.* Is the bound of Remark 5.3 for $H(R_u|R_v)$ sharp, i.e. does $H(R_u|R_v) = H(\Delta_2 | u^* v \Delta_2 v^* u)$ hold?

(8) *Non-abelian subalgebras.* Explicit formulas for the examples of Remark 6.4, starting with $M_2 \otimes 1 \subset M_4$.

A Numerical method

All computations for Table 1 (fourth column), Table 2, Figure 4 and Remark 6.4 use our PyTorch [16] library `torch_vn_algebra` [11]; the program is `scripts/connes_stormer.py` of the repository [11], run in double precision (complex128) on a CPU (full run on CPU (4 threads), 726 s in total). The curves of Figures 1–3 and the third and fifth columns of Table 1 are computed directly from the formulas of Section 4 with NumPy [9] and SciPy [24] (`scripts/connes_stormer_figures.py` for the figures).

The functional. Δ_n is the image of the inclusion $\mathbb{C}^n \subset M_n$ with inclusion matrix $(1, \ldots, 1)$, and Q_U the image of the same inclusion twisted by U (`inclusions.Inclusion` with `unitary=U`); the conditional expectations onto both are taken for the normalised trace $\tau = \mathrm{Tr}/n$, and $\tau\eta(E(x))$ is computed from the eigenvalues of $E(x)$. The computation is batched and differentiable. The test suite checks it against the plain NumPy formula (5) to 10^{-12} for random partitions with $n = 2, 3, 5$.

Search spaces. (1) General partitions with $m = n^2$ elements,

$$x_j = S^{-1/2} Z_j Z_j^* S^{-1/2}, \qquad S = \sum_{j=1}^{m} Z_j Z_j^*,$$

with free complex $Z_j \in M_n$; every family with invertible S is a partition of unity, and every partition of unity with m elements arises this way ($Z_j = x_j^{1/2}$, $S = 1$). (2) Orthonormal bases, $x_j = w_j w_j^*$ for the columns w_j of a unitary $W = W_0 \exp(X - X^*)$, with value (6). (3) States $\rho = ZZ^*/\operatorname{Tr}(ZZ^*)$ for G^*; the reported G^* is the larger of this maximum and the values $G(\rho_j)$ at the normalised elements of the best partition found.

Optimiser. `optimize.extremize` runs Adam with the learning rate $0.05 \cdot 0.6^r$ in rounds $r = 0, \dots, 7$, re-centring the unitary chart ($W_0 \leftarrow W$, $X \leftarrow 0$) after every round, and returns per start the best value attained at any iterate; every iterate is an admissible point. All starts for all unitaries form one batch. We used 1200 steps for $n = 3, 4$ and 2000 for $n = 5$, and the numbers of starts per unitary given in Table 2 (16 for the states). The unitaries U are Haar-random, from the library's sampler (QR decomposition with phase correction [13]), with fixed seeds. For $n = 2$ (Table 1) the search over partitions with four elements uses $U = D_1 R_\theta D_2$ with random phases and 16 starts.

Extended basis search. A unitary is flagged if the best partition beats the best basis by more than 1×10^{-8}. For every flagged unitary the basis search is repeated with 64 further starts, half of them Haar-random and half at bases built from the directions of the best partition (the principal eigenvectors of its elements, n of them drawn with probabilities proportional to their traces, perturbed by complex Gaussian noise of size 10^{-3} and orthonormalised by the polar decomposition). For the unitary with the largest remaining gap (for each n) another 1024 starts are run, again half of each kind.

Recomputation in high precision. For every unitary that remains flagged, the best partition and the best basis are recomputed in mpmath with 50 digits: U is replaced by its polar unitarisation $U' = U(U^*U)^{-1/2}$, exactly unitary at this precision, and the partition $x_j = S^{-1/2} Z_j Z_j^* S^{-1/2}$ is re-normalised from the stored Z_j (for the basis, $Z_j = w_j$), so that $\| \sum_j x_j - 1 \|$ is of the order 10^{-50}. This is a high-precision evaluation, not interval arithmetic. The flagged unitaries are stored in the file `connes_stormer.json` written by the program.

Structure of a partition. Elements of trace below 10^{-7} times the largest trace are discarded; an element counts as rank one if its second eigenvalue is at most 10^{-6} times its trace; the principal eigenvectors d, d' of two elements are merged into one direction if $|\langle d, d' \rangle| > 1 - 10^{-4}$.

Checks. For $n = 2$ the search over general partitions with four elements reproduces Theorem 4.4 to 5×10^{-13}, and the optimal basis of Theorem 4.4(b), evaluated in NumPy, gives the formula to 4×10^{-16} (Table 1). The test suite `tests/test_connes_stormer.py` checks $\kappa = 1.3254868$ to 10^{-6}, compares the formula of Theorem 4.4 with a plain BFGS optimisation (SciPy) over partitions of M_2 with three elements for four values of c, checks that 300 random partitions with two to five elements never exceed it, that the value does not depend on the phases of U, the integral representation of φ by quadrature, the Bloch reduction of Lemma 4.3, the ensemble identity of Proposition 2.3(a) for $n = 2, 3, 4$, and the agreement of the mpmath recomputation with the double-precision value. For complementary masas (Fourier matrices, $n = 2, 3, 4$) the search gives $H = \log n$ to 1×10^{-10} (Remark 3.3(ii)), and $H(\Delta_3 | \Delta_3)$ vanishes to 1×10^{-17}.

Code availability. The library, the programs that compute all numbers, tables and figures, and the tests are available at `https://github.com/andre-a-no/von_neumann_type_i` (MIT licence).

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